
NVIDIA
Hands on Lab 3
Dynamo + KV Cache Profiling

LLM Serving Configuration

Hardware

- DGX H100 based cluster
- InfiniBand (Quantum) network
- Ethernet (Spectrum-X) network
- DDN Exascaler storage

Software

- vLLM Inference engine
- Qwen3-0.6B model
- Dynamo + NIXL + KVCR

Configuration

- Slurm based cluster
- Frontend, prefill and decode on one node
- 8 workers
- One GPU per worker

Test

- Send 48 Inference requests
- 2 Concurrent queries at a time

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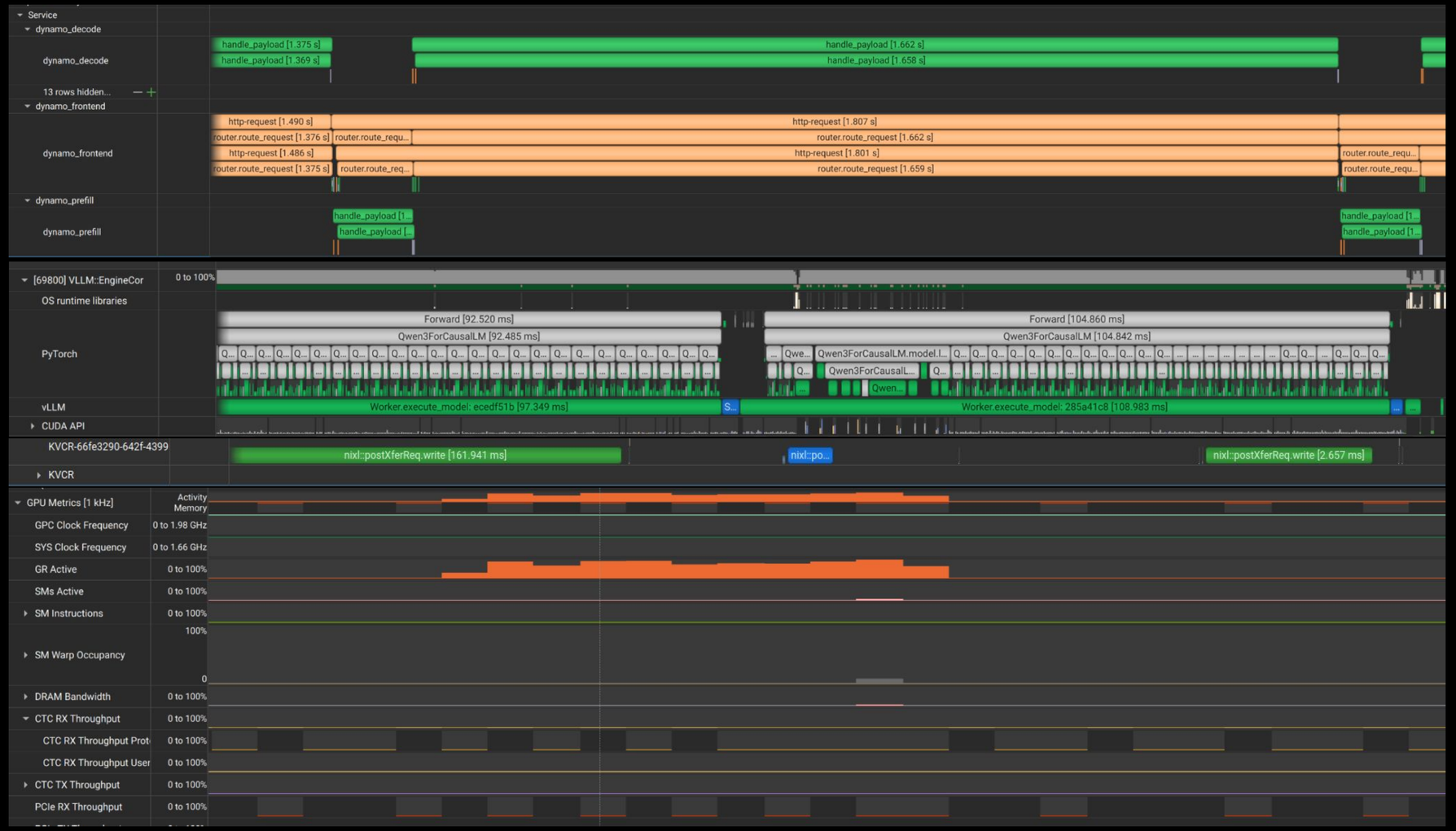
Single node observability

Dynamo

vLLM

KVCR & NIXL

GPU

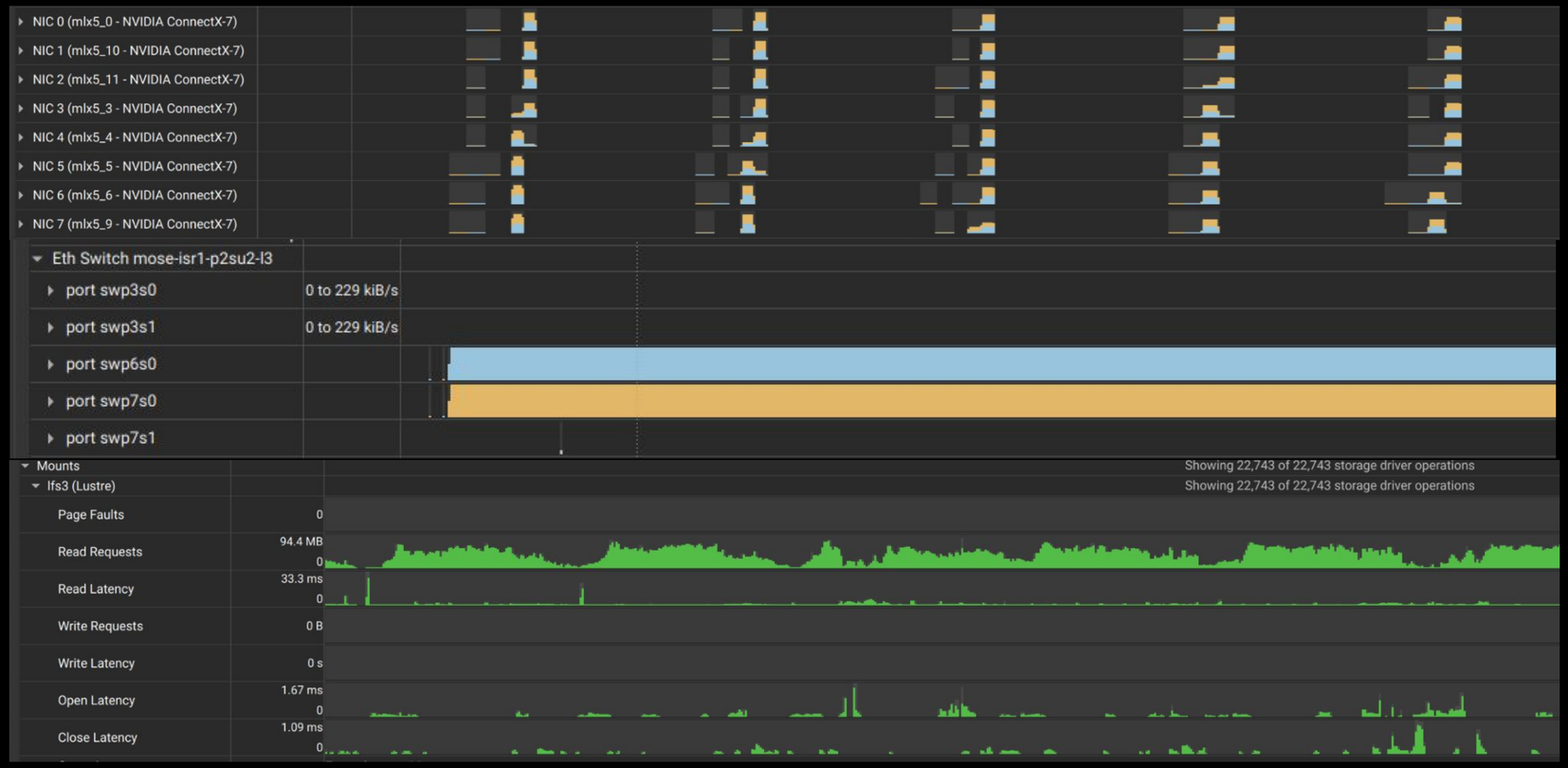


Single node observability /2

NIC

Ethernet Switch

Storage Volume



Nsight Cloud

Cloud/Cluster native components that enable Nsight tools profile workloads in

- Containerized clouds
- Slurm based clusters
- Other data center types

Key features

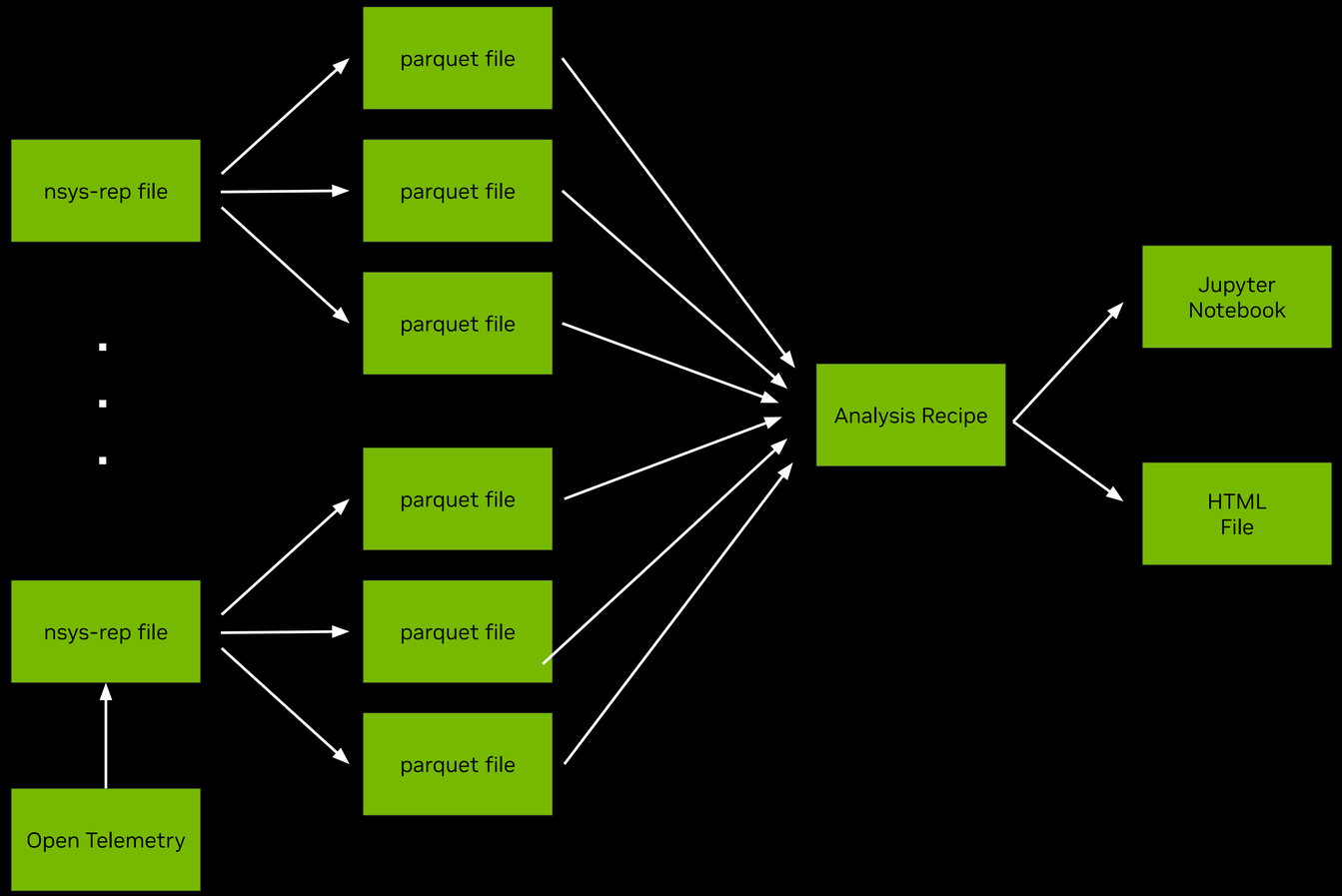
- Profiling K8 containers using Nsight Operator
- Coordinate profiling data acquisition across pods
- Analyze profiled data as a service

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Post Collection Analysis Recipe

A Python script that

- Inputs profile report files collected over multiple nodes
- Exports contained data to parquet
- Reads the parquet tables into python data structures
- Analyzes the data
- Creates HTML or Jupyter notebook files displaying the analysis results



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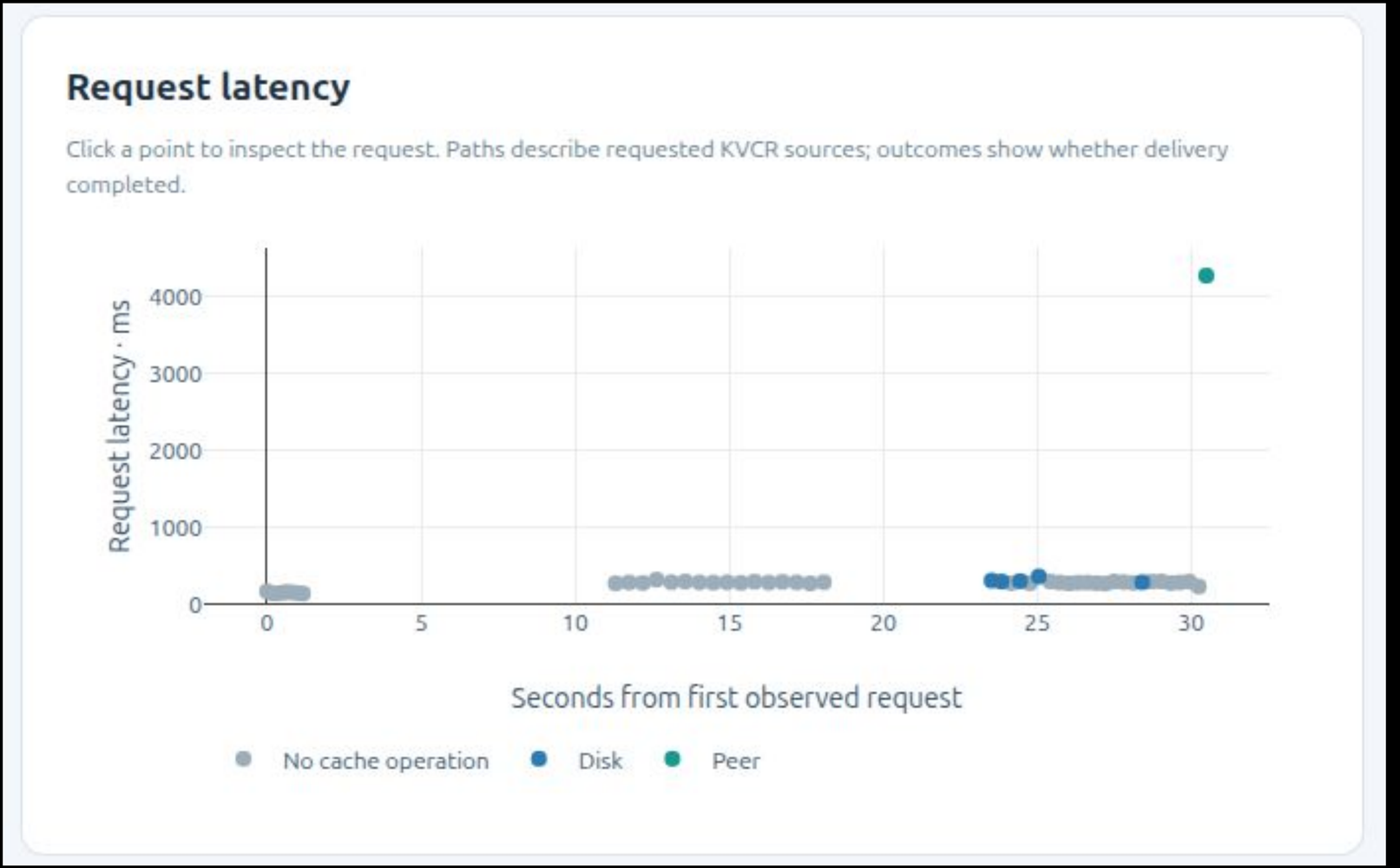
Hands on lab

Please load in a browser:

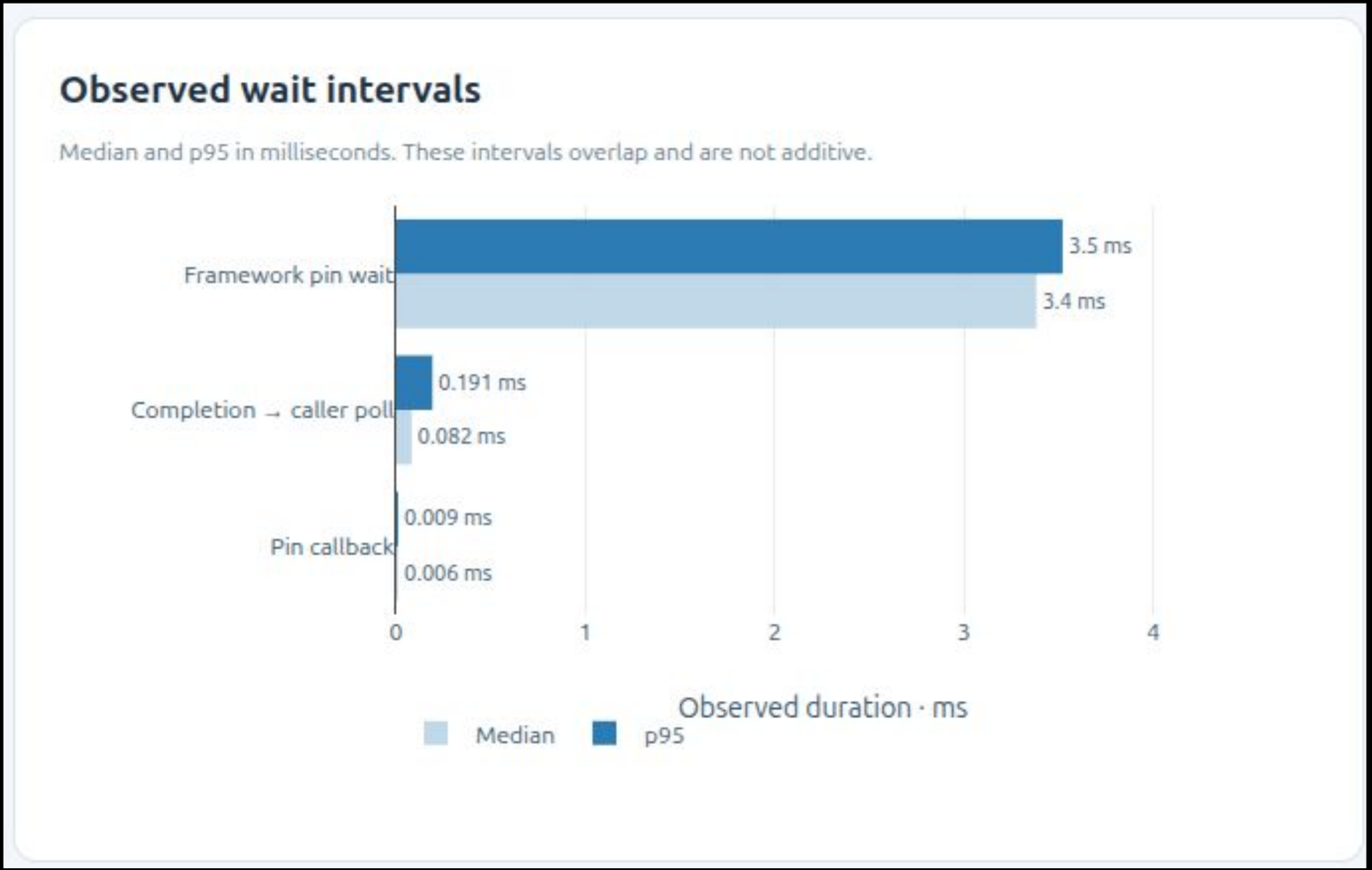
HandsOnLab3/DynamoProfileSession/analysis_reports/kvcr_dashboard.html

And try to locate the AI serving performance problem.

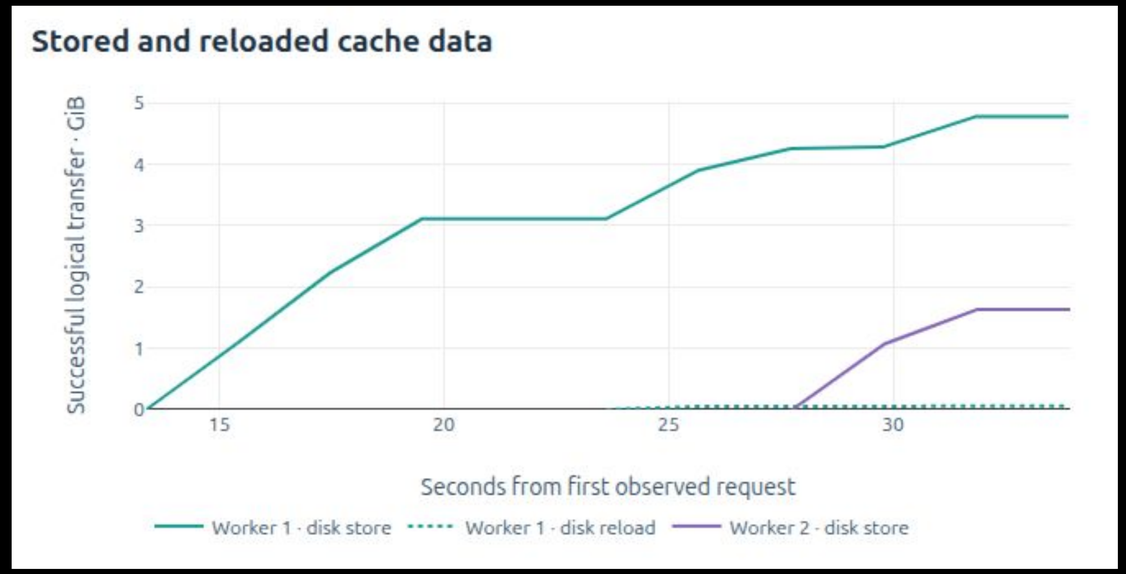
Request Latency

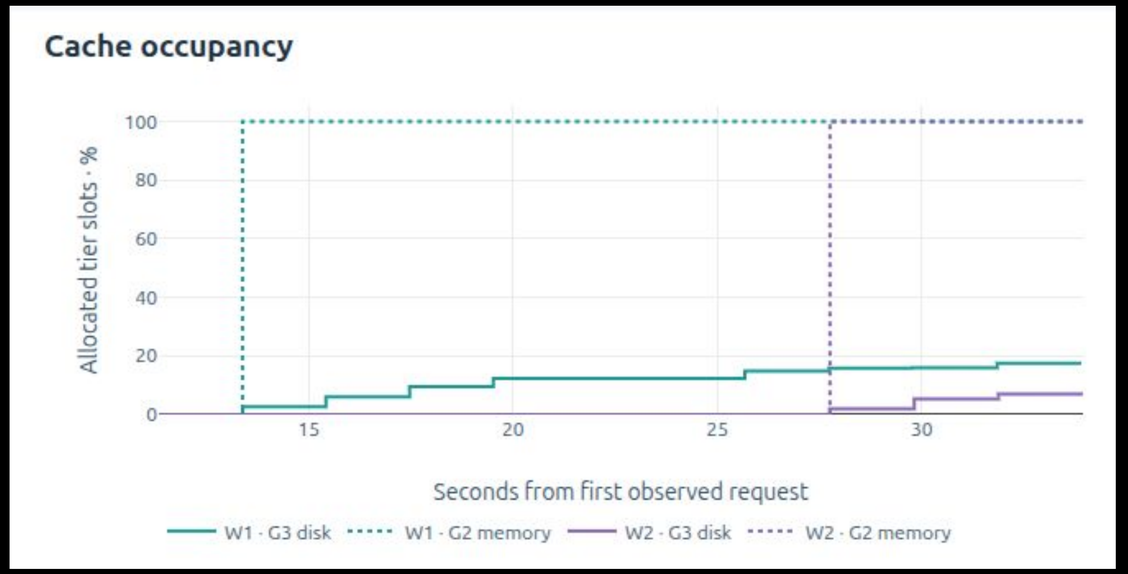


Wait Interval

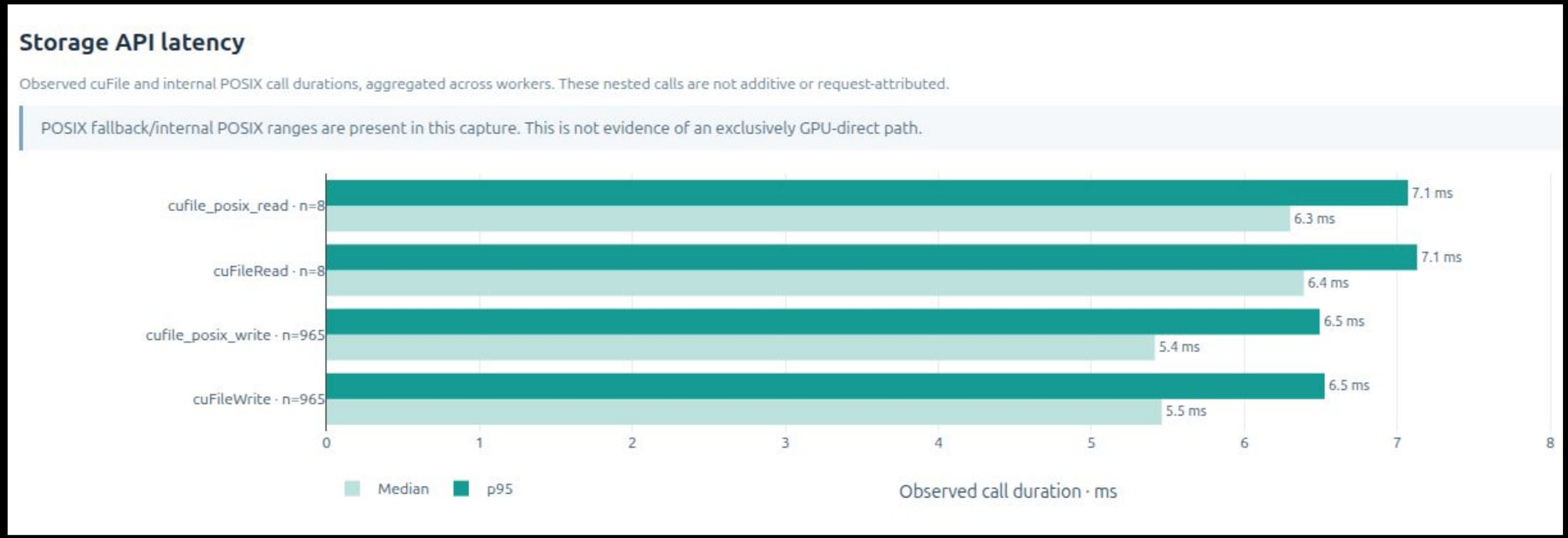


Cache usage and occupancy





Storage API Latency



Storage API Latency is high since GPU Direct Storage isn't configured on the nodes!

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Check GPU Direct Storage Status

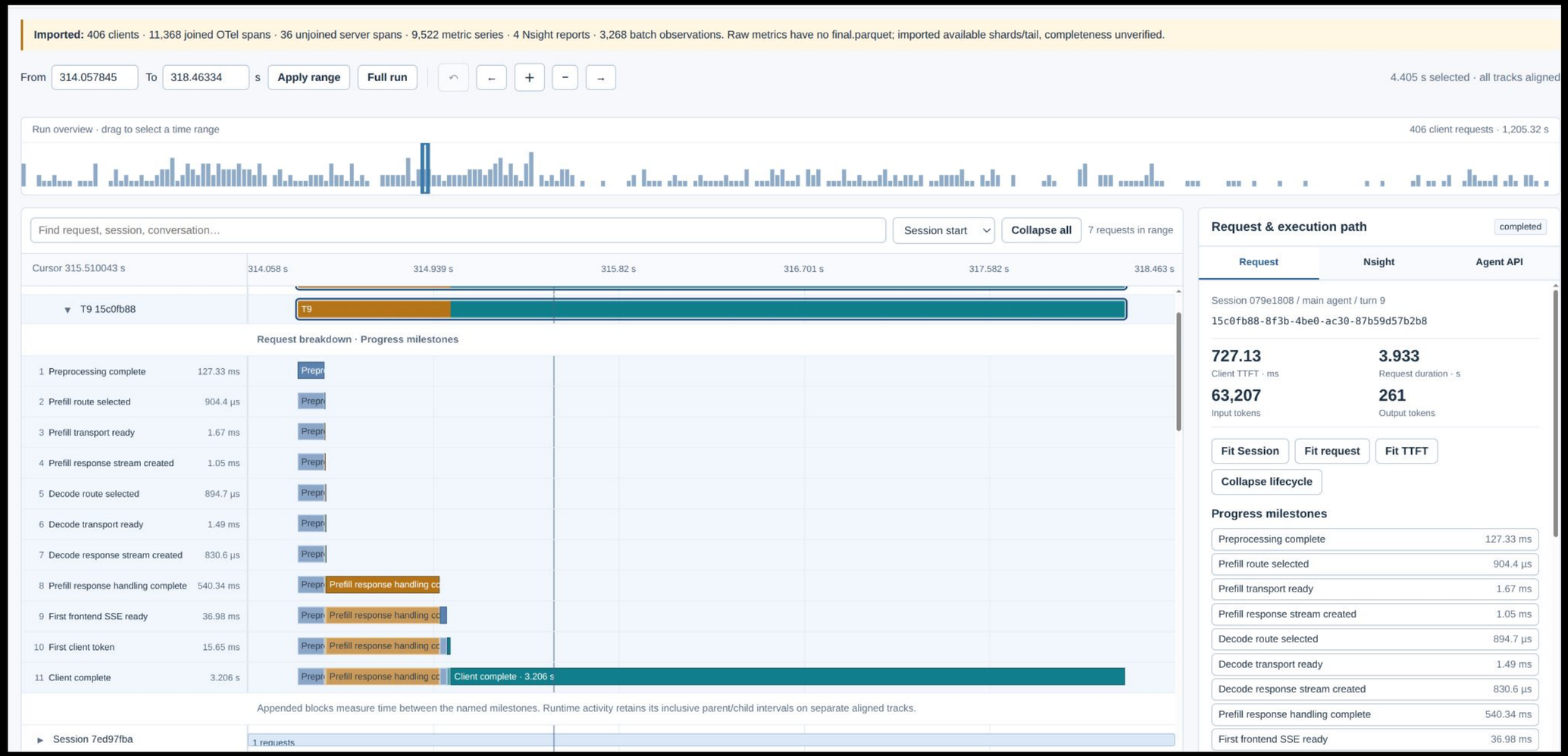
```
root@dt-dgx-h100-1:/usr/local/cuda/gds/tools# ./gdscheck.py -p
 GDS release version: 1.18.1.6
 nvidia_fs minimum version: 2.12
 Platform: x86_64
 ============
 ENVIRONMENT:
 ============
 =====================
 DRIVER CONFIGURATION:
 =====================
 NVMe                : compat
 NVMeOF              : compat
 SCSI                : compat
 ScaleFlux CSD       : compat
 NVMesh              : compat
 DDN EXAScaler       : compat
 IBM Spectrum Scale  : compat
 NFS                 : compat
 BeeGFS              : compat
 ScaTeFS             : compat
 VIRTIOFS            : compat
 WekaFS              : compat
 Userspace RDMA      : Unsupported
 --Mellanox PeerDirect       : Enabled
 --rdma library              : Not Loaded (libcufile_rdma.so)
```

GPU Direct Storage is working in compatibility mode, since libcufile was not installed on the machine.

Compatibility mode = a fallback where data is copied to host memory and then to GPU HBM.

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Dynamo Dsight Explorer



A part of https://github.com/NVIDIA/srt-slurm

KV-Cache usage Best Practices

- Use the right network technology
- Use the right storage technology
- Make sure that GPU Direct Storage is on
- Make sure you have enough KV-Cache storage across all tiers

  GPU HBM -> Host RAM -> Storage
- Configure prefill and decode pools + routing correctly
- Prefer using KVCR (KV Cache Runner)
- Run benchmarks to check how your setup performs
- Profile your workload to pinpoint performance bottlenecks
- Use Monitoring systems to ensure performance doesn't degrade

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NVIDIA Minerva

Minerva is an AI-native AIOps end-to-end observability platform

- Full-stack Observability: collects and correlates telemetry data across GPUs, NVLink switches, Spectrum-X Ethernet networking, storage systems, job schedulers (such as Slurm and Run:ai), and active AI workloads.
- Always-on anomaly detection + predictive maintenance + full stack correlation + AI debugging
- AI assistant that allows investigating cluster issues using natural-language queries.

